

* Rank 0 tensor.

$$[G]$$

$$G = G_s^{(1)}$$

* Rank 1 tensor.

$$\begin{bmatrix} 0 & G_y & 0 \end{bmatrix}$$

$$G_y = G_{py}^{(1)}$$

* Rank 2 tensor (s).

$$\begin{bmatrix} G_{xx} & 0 & G_{xz} \\ 0 & G_{yy} & 0 \\ G_{xz} & 0 & G_{zz} \end{bmatrix}$$

$$G_{xx} = -G_{du}^{(1)} + G_{dv}^{(1)} + G_s^{(1)}$$

$$G_{xz} = G_{dxz}^{(1)}$$

$$G_{yy} = -G_{du}^{(1)} - G_{dv}^{(1)} + G_s^{(1)}$$

$$G_{zz} = 2G_{du}^{(1)} + G_s^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & 0 & G_{xz} \\ 0 & 0 & 0 \\ -G_{xz} & 0 & 0 \end{bmatrix}$$

$$G_{xz} = -Q_{py}^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & G_{xxy} & 0 \\ 0 & G_{yyy} & 0 \\ 0 & G_{zzy} & 0 \\ G_{yzx} & 0 & G_{zzy} \\ 0 & G_{yzx} & 0 \\ G_{xxy} & 0 & G_{yzx} \end{bmatrix}$$

$$G_{xxy} = -G_{fay}^{(1)} - G_{fby}^{(1)} + G_{py}^{(1)}$$

$$G_{yyy} = 2G_{fay}^{(1)} + 3G_{py}^{(1)}$$

$$G_{zzy} = -G_{fay}^{(1)} + G_{fby}^{(1)} + G_{py}^{(1)}$$

$$G_{yzx} = G_{f3}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & G_{xxy} & 0 \\ 0 & 0 & 0 \\ 0 & G_{zzy} & 0 \\ G_{yzx} & 0 & -\frac{G_{zzy}}{2} \\ 0 & G_{zzy} & 0 \\ -\frac{G_{xxy}}{2} & 0 & -G_{yzx} - G_{zzy} \end{bmatrix}$$

$$G_{xxy} = 2G_{py}^{(2)} - 2Q_{dxz}^{(1)}$$

$$G_{zzy} = 2G_{py}^{(2)} + 2Q_{dxz}^{(1)}$$

$$G_{yzx} = 3Q_{du}^{(1)} + Q_{dv}^{(1)}$$

$$\frac{G_{zzy}}{2} = G_{py}^{(2)} + Q_{dxz}^{(1)}$$

$$G_{zxy} = -3Q_{du}^{(1)} + Q_{dv}^{(1)}$$

$$\frac{G_{xxy}}{2} = G_{py}^{(2)} - Q_{dxz}^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} G_{yzx} & 0 & G_{yzz} \\ 0 & G_{zxy} & 0 \\ G_{xyx} & 0 & G_{xyz} \end{bmatrix}$$

$$G_{yzx} = -Q_{du}^{(2)} + Q_{dv}^{(2)} + Q_s^{(1)}$$

$$G_{yzz} = -G_{py}^{(3)} + Q_{dxz}^{(2)}$$

$$G_{zxy} = -Q_{du}^{(2)} - Q_{dv}^{(2)} + Q_s^{(1)}$$

$$G_{xyx} = G_{py}^{(3)} + Q_{dxz}^{(2)}$$

$$G_{xyz} = 2Q_{du}^{(2)} + Q_s^{(1)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} G_{xxxx} & G_{xxyy} & G_{xxzz} & 0 & G_{xxzx} & 0 \\ G_{xxyy} & G_{yyyy} & G_{yyzz} & 0 & G_{yyzx} & 0 \\ G_{xxzz} & G_{yyzz} & G_{zzzz} & 0 & G_{zzzx} & 0 \\ 0 & 0 & 0 & G_{yzyz} & 0 & G_{yzxy} \\ G_{xxzx} & G_{yyzx} & G_{zzzx} & 0 & G_{zxzx} & 0 \\ 0 & 0 & 0 & G_{yzxy} & 0 & G_{xyxy} \end{bmatrix}$$

$$G_{xxxx} = -2G_{du}^{(1)} - 4G_{du}^{(2)} + 2G_{dv}^{(1)} + 4G_{dv}^{(2)} - G_{gu}^{(1)} + G_{gv}^{(1)} + 2G_g^{(1)} + G_s^{(1)} + 2G_s^{(2)}$$

$$G_{xxyy} = -2G_{du}^{(1)} + 2G_{gu}^{(1)} - G_g^{(1)} + G_s^{(1)}$$

$$G_{xxzz} = G_{du}^{(1)} + G_{dv}^{(1)} - G_{gu}^{(1)} - G_{gv}^{(1)} - G_g^{(1)} + G_s^{(1)}$$

$$G_{xxzx} = G_{dxz}^{(1)} + 2G_{dxz}^{(2)} - G_{gay}^{(1)} - G_{gby}^{(1)}$$

$$G_{yyyy} = -2G_{du}^{(1)} - 4G_{du}^{(2)} - 2G_{dv}^{(1)} - 4G_{dv}^{(2)} - G_{gu}^{(1)} - G_{gv}^{(1)} + 2G_g^{(1)} + G_s^{(1)} + 2G_s^{(2)}$$

$$G_{yyzz} = G_{du}^{(1)} - G_{dv}^{(1)} - G_{gu}^{(1)} + G_{gv}^{(1)} - G_g^{(1)} + G_s^{(1)}$$

$$G_{yyzx} = G_{dxz}^{(1)} + 2G_{gby}^{(1)}$$

$$G_{zzzz} = 4G_{du}^{(1)} + 8G_{du}^{(2)} + 2G_{gu}^{(1)} + 2G_g^{(1)} + G_s^{(1)} + 2G_s^{(2)}$$

$$G_{zzzx} = G_{dxz}^{(1)} + 2G_{dxz}^{(2)} + G_{gay}^{(1)} - G_{gby}^{(1)}$$

$$G_{yzyz} = G_{du}^{(2)} - G_{dv}^{(2)} - G_{gu}^{(1)} + G_{gv}^{(1)} - G_g^{(1)} + G_s^{(2)}$$

$$G_{yzxy} = G_{dxz}^{(2)} + 2G_{gby}^{(1)}$$

$$G_{zxzx} = G_{du}^{(2)} + G_{dv}^{(2)} - G_{gu}^{(1)} - G_{gv}^{(1)} - G_g^{(1)} + G_s^{(2)}$$

$$G_{xyxy} = -2G_{du}^{(2)} + 2G_{gu}^{(1)} - G_g^{(1)} + G_s^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & G_{xxyy} & G_{xxzz} & 0 & G_{xxzx} & 0 \\ -G_{xxyy} & 0 & G_{yyzz} & 0 & G_{yyzx} & 0 \\ -G_{xxzz} & -G_{yyzz} & 0 & 0 & G_{zzzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & G_{yzxy} \\ -G_{xxzx} & -G_{yyzx} & -G_{zzzx} & 0 & 0 & 0 \\ 0 & 0 & 0 & -G_{yzxy} & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
G_{xxyy} &= -2G_{dv}^{(3)} + 4Q_{f3}^{(1)} \\
G_{xxzz} &= 3G_{du}^{(3)} - G_{dv}^{(3)} - 4Q_{f3}^{(1)} \\
G_{xxzx} &= G_{dxz}^{(3)} + 2Q_{fay}^{(1)} + 2Q_{fby}^{(1)} - 2Q_{py}^{(1)} \\
G_{yyzz} &= 3G_{du}^{(3)} + G_{dv}^{(3)} + 4Q_{f3}^{(1)} \\
G_{yyzx} &= G_{dxz}^{(3)} - 4Q_{fby}^{(1)} \\
G_{zzzx} &= G_{dxz}^{(3)} - 2Q_{fay}^{(1)} + 2Q_{fby}^{(1)} + 2Q_{py}^{(1)} \\
G_{yzxy} &= 4Q_{fay}^{(1)} + Q_{py}^{(1)}
\end{aligned}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} G_{yzyz} & 0 & G_{yzxy} \\ 0 & G_{zxzx} & 0 \\ G_{yzxy} & 0 & G_{xyxy} \end{bmatrix}$$

$$\begin{aligned}
G_{yzyz} &= -2G_{du}^{(4)} + 2G_{dv}^{(4)} + G_s^{(3)} \\
G_{yzxy} &= 2G_{dxz}^{(4)} \\
G_{zxzx} &= -2G_{du}^{(4)} - 2G_{dv}^{(4)} + G_s^{(3)} \\
G_{xyxy} &= 4G_{du}^{(4)} + G_s^{(3)}
\end{aligned}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & 0 & G_{yzxy} \\ 0 & 0 & 0 \\ -G_{yzxy} & 0 & 0 \end{bmatrix}$$

$$G_{yzxy} = Q_{py}^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & G_{xxzx} & 0 \\ 0 & G_{yyzx} & 0 \\ 0 & G_{zzzx} & 0 \\ G_{yzyz} & 0 & G_{yzxy} \\ 0 & G_{zxzx} & 0 \\ G_{xyyz} & 0 & G_{xyxy} \end{bmatrix}$$

$$\begin{aligned}
G_{xxzx} &= 2G_{dxz}^{(5)} - Q_{fay}^{(2)} - Q_{fby}^{(2)} + Q_{py}^{(3)} \\
G_{yyzx} &= 2Q_{fay}^{(2)} + Q_{py}^{(3)} + 2Q_{py}^{(4)} \\
G_{zzzx} &= -2G_{dxz}^{(5)} - Q_{fay}^{(2)} + Q_{fby}^{(2)} + Q_{py}^{(3)} \\
G_{yzyz} &= -3G_{du}^{(5)} - G_{dv}^{(5)} + Q_{f3}^{(2)} \\
G_{yzxy} &= G_{dxz}^{(5)} - Q_{fay}^{(2)} + Q_{fby}^{(2)} + Q_{py}^{(4)} \\
G_{zxzx} &= 3G_{du}^{(5)} - G_{dv}^{(5)} + Q_{f3}^{(2)} \\
G_{xyyz} &= -G_{dxz}^{(5)} - Q_{fay}^{(2)} - Q_{fby}^{(2)} + Q_{py}^{(4)} \\
G_{xyxy} &= 2G_{dv}^{(5)} + Q_{f3}^{(2)}
\end{aligned}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & G_{yzyz} & 0 & G_{yzxy} \\ G_{zxzx} & G_{zyyz} & G_{zxzx} & 0 & G_{zxzx} & 0 \\ 0 & 0 & 0 & G_{xyyz} & 0 & G_{xyxy} \end{bmatrix}$$

$$\begin{aligned}
G_{yzyz} &= -3G_{du}^{(6)} - G_{dv}^{(6)} + Q_{f3}^{(3)} \\
G_{yzxy} &= -G_{dxz}^{(6)} - Q_{fay}^{(3)} - Q_{fby}^{(3)} + Q_{py}^{(6)} \\
G_{zxxx} &= 2G_{dxz}^{(6)} - Q_{fay}^{(3)} - Q_{fby}^{(3)} + Q_{py}^{(5)} \\
G_{zxyy} &= 2Q_{fay}^{(3)} + Q_{py}^{(5)} + 2Q_{py}^{(6)} \\
G_{zxxx} &= -2G_{dxz}^{(6)} - Q_{fay}^{(3)} + Q_{fby}^{(3)} + Q_{py}^{(5)} \\
G_{zxxx} &= 3G_{du}^{(6)} - G_{dv}^{(6)} + Q_{f3}^{(3)} \\
G_{xyyz} &= G_{dxz}^{(6)} - Q_{fay}^{(3)} + Q_{fby}^{(3)} + Q_{py}^{(6)} \\
G_{xyxy} &= 2G_{dv}^{(6)} + Q_{f3}^{(3)}
\end{aligned}$$

* Rank 4 tensor (s).

$$\begin{bmatrix}
G_{xxxx} & G_{xxyy} & G_{xxzz} & 0 & G_{xxxz} & 0 & 0 & G_{xxxz} & 0 \\
G_{yyxx} & G_{yyyy} & G_{yyzz} & 0 & G_{yyzx} & 0 & 0 & G_{yyxz} & 0 \\
G_{zzxx} & G_{zzyy} & G_{zzzz} & 0 & G_{zzzx} & 0 & 0 & G_{zzxz} & 0 \\
0 & 0 & 0 & G_{yzyz} & 0 & G_{yzxy} & G_{yzzz} & 0 & G_{yzyx} \\
G_{zxxx} & G_{zxyy} & G_{zxzz} & 0 & G_{zxzx} & 0 & 0 & G_{zxxx} & 0 \\
0 & 0 & 0 & G_{xyyz} & 0 & G_{xyxy} & G_{xyzy} & 0 & G_{xyyx}
\end{bmatrix}$$

$$\begin{aligned}
G_{xxxx} &= -2G_{du}^{(1)} - 4G_{du}^{(2)} + 2G_{dv}^{(1)} + 4G_{dv}^{(2)} - G_{gu}^{(1)} + G_{gv}^{(1)} + 2G_g^{(1)} + G_s^{(1)} + 2G_s^{(2)} \\
G_{xxyy} &= -2G_{du}^{(1)} - 2G_{dv}^{(3)} + 2G_{gu}^{(1)} - G_g^{(1)} + G_s^{(1)} + 4Q_{f3}^{(1)} \\
G_{xxzz} &= G_{du}^{(1)} + 3G_{du}^{(3)} + G_{dv}^{(1)} - G_{dv}^{(3)} - G_{gu}^{(1)} - G_{gv}^{(1)} - G_g^{(1)} + G_s^{(1)} - 4Q_{f3}^{(1)} \\
G_{xxzx} &= G_{dxz}^{(1)} + 2G_{dxz}^{(2)} + G_{dxz}^{(3)} + 2G_{dxz}^{(5)} - G_{gay}^{(1)} - G_{gby}^{(1)} + 2Q_{fay}^{(1)} - Q_{fay}^{(2)} + 2Q_{fby}^{(1)} - Q_{fby}^{(2)} - 2Q_{py}^{(1)} + Q_{py}^{(3)} \\
G_{xxxz} &= G_{dxz}^{(1)} + 2G_{dxz}^{(2)} + G_{dxz}^{(3)} - 2G_{dxz}^{(5)} - G_{gay}^{(1)} - G_{gby}^{(1)} + 2Q_{fay}^{(1)} + Q_{fay}^{(2)} + 2Q_{fby}^{(1)} + Q_{fby}^{(2)} - 2Q_{py}^{(1)} - Q_{py}^{(3)} \\
G_{yyxx} &= -2G_{du}^{(1)} + 2G_{dv}^{(3)} + 2G_{gu}^{(1)} - G_g^{(1)} + G_s^{(1)} - 4Q_{f3}^{(1)} \\
G_{yyyy} &= -2G_{du}^{(1)} - 4G_{du}^{(2)} - 2G_{dv}^{(1)} - 4G_{dv}^{(2)} - G_{gu}^{(1)} - G_{gv}^{(1)} + 2G_g^{(1)} + G_s^{(1)} + 2G_s^{(2)} \\
G_{yyzz} &= G_{du}^{(1)} + 3G_{du}^{(3)} - G_{dv}^{(1)} + G_{dv}^{(3)} - G_{gu}^{(1)} + G_{gv}^{(1)} - G_g^{(1)} + G_s^{(1)} + 4Q_{f3}^{(1)} \\
G_{yyzx} &= G_{dxz}^{(1)} + G_{dxz}^{(3)} + 2G_{dxz}^{(1)} + 2Q_{fay}^{(2)} - 4Q_{fby}^{(1)} + Q_{py}^{(3)} + 2Q_{py}^{(4)} \\
G_{yyxz} &= G_{dxz}^{(1)} + G_{dxz}^{(3)} + 2G_{gby}^{(1)} - 2Q_{fay}^{(2)} - 4Q_{fby}^{(1)} - Q_{py}^{(3)} - 2Q_{py}^{(4)} \\
G_{zzxx} &= G_{du}^{(1)} - 3G_{du}^{(3)} + G_{dv}^{(1)} + G_{dv}^{(3)} - G_{gu}^{(1)} - G_{gv}^{(1)} - G_g^{(1)} + G_s^{(1)} + 4Q_{f3}^{(1)} \\
G_{zzyy} &= G_{du}^{(1)} - 3G_{du}^{(3)} - G_{dv}^{(1)} - G_{dv}^{(3)} - G_{gu}^{(1)} + G_{gv}^{(1)} - G_g^{(1)} + G_s^{(1)} - 4Q_{f3}^{(1)} \\
G_{zzzz} &= 4G_{du}^{(1)} + 8G_{du}^{(2)} + 2G_{gu}^{(1)} + 2G_g^{(1)} + G_s^{(1)} + 2G_s^{(2)} \\
G_{zzzx} &= G_{dxz}^{(1)} + 2G_{dxz}^{(2)} + G_{dxz}^{(3)} - 2G_{dxz}^{(5)} + G_{gay}^{(1)} - G_{gby}^{(1)} - 2Q_{fay}^{(1)} - Q_{fay}^{(2)} + 2Q_{fby}^{(1)} + Q_{fby}^{(2)} + 2Q_{py}^{(1)} + Q_{py}^{(3)} \\
G_{zzxz} &= G_{dxz}^{(1)} + 2G_{dxz}^{(2)} + G_{dxz}^{(3)} + 2G_{dxz}^{(5)} + G_{gay}^{(1)} - G_{gby}^{(1)} - 2Q_{fay}^{(1)} + Q_{fay}^{(2)} + 2Q_{fby}^{(1)} - Q_{fby}^{(2)} + 2Q_{py}^{(1)} - Q_{py}^{(3)} \\
G_{yzyz} &= G_{du}^{(2)} - 3G_{du}^{(5)} - G_{dv}^{(2)} - G_{dv}^{(5)} - G_{gu}^{(1)} + G_{gv}^{(1)} - G_g^{(1)} + G_s^{(2)} + Q_{f3}^{(2)} \\
G_{yzxy} &= G_{dxz}^{(2)} + G_{dxz}^{(5)} + 2G_{gby}^{(1)} + 4Q_{fay}^{(1)} - Q_{fay}^{(2)} + Q_{fby}^{(2)} + Q_{py}^{(1)} + Q_{py}^{(4)} \\
G_{yzzy} &= G_{du}^{(2)} + 3G_{du}^{(5)} - G_{dv}^{(2)} + G_{dv}^{(5)} - G_{gu}^{(1)} + G_{gv}^{(1)} - G_g^{(1)} + G_s^{(2)} - Q_{f3}^{(2)} \\
G_{yzyx} &= G_{dxz}^{(2)} - G_{dxz}^{(5)} + 2G_{gby}^{(1)} + 4Q_{fay}^{(1)} + Q_{fay}^{(2)} - Q_{fby}^{(2)} + Q_{py}^{(1)} - Q_{py}^{(4)} \\
G_{zxxx} &= G_{dxz}^{(1)} + 2G_{dxz}^{(2)} - G_{dxz}^{(3)} - G_{gay}^{(1)} - G_{gby}^{(1)} - 2Q_{fay}^{(1)} - 2Q_{fby}^{(1)} + 2Q_{py}^{(1)} \\
G_{zxyy} &= G_{dxz}^{(1)} - G_{dxz}^{(3)} + 2G_{gby}^{(1)} + 4Q_{fby}^{(1)} \\
G_{zzzz} &= G_{dxz}^{(1)} + 2G_{dxz}^{(2)} - G_{dxz}^{(3)} + G_{gay}^{(1)} - G_{gby}^{(1)} + 2Q_{fay}^{(1)} - 2Q_{fby}^{(1)} - 2Q_{py}^{(1)} \\
G_{zzzx} &= G_{du}^{(2)} + 3G_{du}^{(5)} + G_{dv}^{(2)} - G_{dv}^{(5)} - G_{gu}^{(1)} - G_{gv}^{(1)} - G_g^{(1)} + G_s^{(2)} + Q_{f3}^{(2)} \\
G_{zzxz} &= G_{du}^{(2)} - 3G_{du}^{(5)} + G_{dv}^{(2)} + G_{dv}^{(5)} - G_{gu}^{(1)} - G_{gv}^{(1)} - G_g^{(1)} + G_s^{(2)} - Q_{f3}^{(2)} \\
G_{xyyz} &= G_{dxz}^{(2)} - G_{dxz}^{(5)} + 2G_{gby}^{(1)} - 4Q_{fay}^{(1)} - Q_{fay}^{(2)} - Q_{fby}^{(2)} - Q_{py}^{(1)} + Q_{py}^{(4)} \\
G_{xyxy} &= -2G_{du}^{(2)} + 2G_{dv}^{(5)} + 2G_{gu}^{(1)} - G_g^{(1)} + G_s^{(2)} + Q_{f3}^{(2)} \\
G_{xyzy} &= G_{dxz}^{(2)} + G_{dxz}^{(5)} + 2G_{gby}^{(1)} - 4Q_{fay}^{(1)} + Q_{fay}^{(2)} + Q_{fby}^{(2)} - Q_{py}^{(1)} - Q_{py}^{(4)} \\
G_{xyyx} &= -2G_{du}^{(2)} - 2G_{dv}^{(5)} + 2G_{gu}^{(1)} - G_g^{(1)} + G_s^{(2)} - Q_{f3}^{(2)}
\end{aligned}$$

* Rank 4 tensor (a).

$$\begin{bmatrix}
0 & 0 & 0 & G_{yzyz} & 0 & G_{yzxy} & G_{yzzy} & 0 & G_{yzyx} \\
G_{zxxx} & G_{zxyy} & G_{zxxx} & 0 & G_{zxxx} & 0 & 0 & G_{zxxx} & 0 \\
0 & 0 & 0 & G_{xyyz} & 0 & G_{xyxy} & G_{xyzy} & 0 & G_{xyyx}
\end{bmatrix}$$

$$\begin{aligned}
G_{yzyz} &= -2G_{du}^{(4)} - 3G_{du}^{(6)} + 2G_{dv}^{(4)} - G_{dv}^{(6)} + G_s^{(3)} + Q_{f3}^{(3)} \\
G_{yzxy} &= 2G_{dxz}^{(4)} - G_{dxz}^{(6)} - Q_{fay}^{(3)} - Q_{fby}^{(3)} + Q_{py}^{(2)} + Q_{py}^{(6)} \\
G_{yzzz} &= 2G_{du}^{(4)} - 3G_{du}^{(6)} - 2G_{dv}^{(4)} - G_{dv}^{(6)} - G_s^{(3)} + Q_{f3}^{(3)} \\
G_{yzyx} &= -2G_{dxz}^{(4)} - G_{dxz}^{(6)} - Q_{fay}^{(3)} - Q_{fby}^{(3)} - Q_{py}^{(2)} + Q_{py}^{(6)} \\
G_{zxxx} &= 2G_{dxz}^{(6)} - Q_{fay}^{(3)} - Q_{fby}^{(3)} + Q_{py}^{(5)} \\
G_{zxyy} &= 2Q_{fay}^{(3)} + Q_{py}^{(5)} + 2Q_{py}^{(6)} \\
G_{zxxz} &= -2G_{dxz}^{(6)} - Q_{fay}^{(3)} + Q_{fby}^{(3)} + Q_{py}^{(5)} \\
G_{zxxz} &= -2G_{du}^{(4)} + 3G_{du}^{(6)} - 2G_{dv}^{(4)} - G_{dv}^{(6)} + G_s^{(3)} + Q_{f3}^{(3)} \\
G_{zxxz} &= 2G_{du}^{(4)} + 3G_{du}^{(6)} + 2G_{dv}^{(4)} - G_{dv}^{(6)} - G_s^{(3)} + Q_{f3}^{(3)} \\
G_{xyyz} &= 2G_{dxz}^{(4)} + G_{dxz}^{(6)} - Q_{fay}^{(3)} + Q_{fby}^{(3)} - Q_{py}^{(2)} + Q_{py}^{(6)} \\
G_{xyxy} &= 4G_{du}^{(4)} + 2G_{dv}^{(6)} + G_s^{(3)} + Q_{f3}^{(3)} \\
G_{xyzy} &= -2G_{dxz}^{(4)} + G_{dxz}^{(6)} - Q_{fay}^{(3)} + Q_{fby}^{(3)} + Q_{py}^{(2)} + Q_{py}^{(6)} \\
G_{xyyx} &= -4G_{du}^{(4)} + 2G_{dv}^{(6)} - G_s^{(3)} + Q_{f3}^{(3)}
\end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix}
G_{xxxx} & 0 & G_{xxxx} \\
0 & G_{yyyy} & 0 \\
G_{zzzx} & 0 & G_{zzzx} \\
G_{yyzx} & 0 & G_{yyzx} \\
G_{zzxx} & 0 & G_{zzxx} \\
0 & G_{xxyy} & 0 \\
0 & G_{yyzz} & 0 \\
G_{zzxx} & 0 & G_{zzxx} \\
G_{xxyy} + G_{yyzz} - G_{yzzy} - G_{zxxz} + G_{zzxx} & 0 & G_{xxyy} \\
0 & G_{xyyz} + \frac{G_{zzzz}}{2} - \frac{G_{zzzz}}{2} & 0
\end{bmatrix}$$

$$\begin{aligned}
G_{xxxx} &= -2G_{du}^{(1)} - 4G_{du}^{(2)} + 2G_{dv}^{(1)} + 4G_{dv}^{(2)} - G_{gu}^{(1)} + G_{gv}^{(1)} + 2G_g^{(1)} + 3G_s^{(1)} \\
G_{xxzx} &= G_{dxz}^{(1)} + 2G_{dxz}^{(2)} + G_{dxz}^{(3)} - 2G_{dxz}^{(5)} - G_{gay}^{(1)} - G_{gby}^{(1)} \\
G_{yyyy} &= -2G_{du}^{(1)} - 4G_{du}^{(2)} - 2G_{dv}^{(1)} - 4G_{dv}^{(2)} - G_{gu}^{(1)} - G_{gv}^{(1)} + 2G_g^{(1)} + 3G_s^{(1)} \\
G_{zzzx} &= G_{dxz}^{(1)} + 2G_{dxz}^{(2)} + G_{dxz}^{(3)} - 2G_{dxz}^{(5)} + G_{gay}^{(1)} - G_{gby}^{(1)} \\
G_{zzzz} &= 4G_{du}^{(1)} + 8G_{du}^{(2)} + 2G_{gu}^{(1)} + 2G_g^{(1)} + 3G_s^{(1)} \\
G_{yyzx} &= G_{dxz}^{(1)} + G_{dxz}^{(3)} + 2G_{gby}^{(1)} \\
G_{yyzz} &= G_{du}^{(1)} + 3G_{du}^{(3)} - G_{dv}^{(1)} + G_{dv}^{(3)} - G_{gu}^{(1)} + G_{gv}^{(1)} - G_g^{(1)} + G_s^{(1)} \\
G_{zzxx} &= G_{du}^{(1)} - 3G_{du}^{(3)} + G_{dv}^{(1)} + G_{dv}^{(3)} - G_{gu}^{(1)} - G_{gv}^{(1)} - G_g^{(1)} + G_s^{(1)} \\
G_{zzzx} &= G_{dxz}^{(1)} + 2G_{dxz}^{(2)} + G_{dxz}^{(3)} + 2G_{dxz}^{(5)} + G_{gay}^{(1)} - G_{gby}^{(1)} \\
G_{xxyy} &= -2G_{du}^{(1)} - 2G_{dv}^{(3)} + 2G_{gu}^{(1)} - G_g^{(1)} + G_s^{(1)} \\
G_{yzyz} &= G_{du}^{(2)} + 3G_{du}^{(5)} - G_{dv}^{(2)} + G_{dv}^{(5)} - G_{gu}^{(1)} + G_{gv}^{(1)} - G_g^{(1)} + G_s^{(1)} \\
G_{zxxx} &= G_{dxz}^{(1)} + 2G_{dxz}^{(2)} - G_{dxz}^{(3)} - G_{gay}^{(1)} - G_{gby}^{(1)} \\
G_{zxxz} &= G_{du}^{(2)} - 3G_{du}^{(5)} + G_{dv}^{(2)} + G_{dv}^{(5)} - G_{gu}^{(1)} - G_{gv}^{(1)} - G_g^{(1)} + G_s^{(1)} \\
G_{xyyz} &= G_{dxz}^{(2)} - G_{dxz}^{(5)} + 2G_{gby}^{(1)}
\end{aligned}$$